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Project Name ; Lanco Hills-LH10 Project Hyderabad		

PROJECT QUALITY PLAN (PQP)

LOCATED AT

**LANCO HILLS -LH10 PROJECT,
LANCO HILLS APARTMENT ,
MANIKONDA, HYDERABAD-500089**

BCIM ENGINEERING PRIVATE LIMITED

**NO -11 B WING, DIVYASREE CHAMBERS
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QUALITY POLICY

We, at **BCIM Engineering Private Limited**, are committed to a high standard of quality practices for all our customers with excellent quality products and service in all our project, which satisfies or exceeds the clients and applicable requirements.

In accomplishing the above, we are committed to

- Ensuring Quality Management System is established, implemented maintain and continually improved in compliance to International Standard.
- Ensuring that all our contracts and project management plans are monitored and managed efficiently and effectively for on time completion and handover of projects.
- Monitoring and evaluating the quality performance of all our employees, subcontractors and suppliers and implementing effective communication with them on quality confirmation and applicable compliance requirement.
- Establishing appropriate responsibility, accountability and authority for quality at all levels.
- Providing a conducive learning environments to facilities sharing, educating and training employees towards increased awareness, continual improvements, knowledge and skill enhancements.

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2. Purpose

This project quality plan describing how BCIM intends to manage the quality issues on this project.

This plan interfaces with the other project plans to form the overall BCIM management system for this project.

This plan is developed to provide adequate guidelines for all parties involved to carry the works specified in the contract in a planned and controlled manner that satisfies contractual and company requirements. The project quality plan also provides a system for managing, monitoring and recording of the performance of the quality system.

The BCIM Project Director is responsible to ensure all planned quality assurance and control activities are implemented and completed by the relevant Project team members.

2.1. Scope

The System for managing quality on this Project is part of the BCIM Management System. The Project Management Plan interfaces closely with the other functional systems on the Project and runs as a single integrated system.

The effect of this integrated approach is that there are many functions of management with impact on quality that are covered in other and more appropriate Project Plans and not duplicated in this Plan (e.g., for procurement activities, it is more appropriate to document in the Procurement Plan).

This Project Quality Plan is important but one of the suites of Project Plans that covers all aspects of quality management on the Project and the requirements of ISO 9001:2015.

Terms/Abbreviations used in this Plan

Env	Environmental
ITP	Inspection and Test Plan
NCR	Non Conformance Report
OHS	Occupational Health and Safety
PQP	Project Quality Plan (this Plan)
QA	Quality Assurance
QC	Quality Control
SOP	Standard Operating Procedures

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TPI Third Party Inspection

WMS Work Method Statement

3. QUALITY TEAM

The Quality Team on the Project is set-up as per overall organization structure for the Project. Only key positions are indicated. In general, the Project Quality Manager is responsible for quality management of the Project.

The positions may change throughout the life of the Project based on situation and work requirements as determined by the Project Director. Such changes are reflected in a

Detailed organization chart which is produced and updated monthly for inclusion into the Monthly Project Progress Reports.

3.1 Responsibilities, Authorities and Functions

Responsibilities of Key Project Team Personnel

The Project Manager is responsible to assign tasks to each member of the Project Team and the limits of the area to be covered by each member. These may change from time to time as decided by the Project Director as the Project progresses. These assignments are reviewed and communicated to the team by the Project Manager during the monthly meetings

3.1.1KEY POSITION AND RESPONSIBILITIES

PROJECT DIRECTOR

Reports to Managing Director and intimating project progress timely basis

Full and final responsibility for the delivery of the Project in accordance with the company's Performance guidelines and the Client expectations

Liaise with the designated liaison manager to ensure they are kept up to date on the issues and progress of the site.

To identify appropriate resources (people, equipment, procedure and processes) in order to provide the facility to implement the works.



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To establish and maintain the physical environment for **EHS (Environment Health, Safety, and)** in order to meet specific group policies.

To manage relationship with client and other key stakeholders in order to foster a

Positive image and potential repeat business

To manage the implementation of the works in order to meet the budgetary and time Expectations of the joint venture

To foster an environment that provides opportunities and enables development, Progression and succession internally, and have an understanding of external (competitors) in order to provide sustainability of the human resource assets in the company

Act decisively in emergency situations

Overall responsibility for the execution of the project in accordance with the requirements of the contract Responsibility to implement and maintain the Project Management Plan for the processes Under his responsibility

Conducts the project meeting as and when required.

Promote an interest and enthusiasm within your required work for health, safety and environment. Self-promote and lead on best practice and act in accordance with the prescribed management system and procedures in fulfilling an injury and incident free working environment

PROJECT MANAGER

Reports to Project Director and intimating project progress on time to time.

Full and final responsibility for the delivery of the Project in accordance with the company's Performance guidelines and the Client expectations as agreed in contract.

Liaise with the designated liaison manager to ensure they are kept up to date on the issues and progress of the site.

To identify appropriate resources (people, equipment, procedure and processes) in order to provide the facility to implement the works.

To establish and maintain the physical environment for HSE (Health, Safety, Environment and) in order to meet specific group policies.

To be responsible for;

- Subcontractors overall performance and progress.

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- Keeping within budgeted costs
- The adequacy of resources employed
- Construction planning and working methods.
- Site safety and compliance with Project Safety Plan and all local regulations
- Quality of works in accordance with the Design and Project Quality Plan and to record non-compliance.

HSE INCHARGE

Advise the Project management team on:

Preventing injury to personnel and damage to plant and equipment. Further improvement in existing sound working methods.

Legal requirements affecting safety, health, welfare and environmental. Provision and use of protective clothing and equipment.

Potential hazards on new contracts before work starts and on the site safety organization.

New developments in methods of safe working.

Carry out site surveys in association with the project management team to ensure that only safe methods of working are in operation and that all regulations are being observed.

Determine the cause of any accident or dangerous occurrence and recommend means of preventing recurrence.

Conduct safety induction for all Project and subcontractors' employees on the Project's OH&S requirements.

Manage/carry out the safety training of all levels of employees on the Project.

Promote awareness of injury prevention and damage control to all levels of employees. Provide feedback on new policies or procedures adopted.

Keep up-to-date with recommended codes of practices and new safety literature and circulate information applicable to each level of staff.

Functional reporting to the Regional Quality and Safety in charge on Safety Performance of the Project to comply with Regional Procedures.

Fulfill the role of Safety Officer required in the OH&S Act.

Supervise the recording and analysis of information on injuries, damage and loss, assess accident trends and review overall safety performance of the project



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Assist the **Project QA Engineer** in carrying out environmental inspections and monitoring. Maintain contact and relationship with officials and professional bodies.

Supervise the work and performance of the Safety Officers and other subordinates, as designated.

Assist in the development and maintenance of Company systems. Other duties as directed

QA/QC INCHARGE

Reports to the Project Director

To liaise with construction team to ensure all inspection and testing requirements are fulfilled and completed work quality is achieved in order to meet client's requirement as per the contract

To liaise with client's representative on site to ensure all inspection and testing requirements are fulfilled and complete work quality is achieved in order to meet client's requirement as per the contract

To ensure all records are completed and compiled for submission to client with regards to completed work quality in order to meet client's requirement as per the contract.

To liaise with construction team to ensure quality of products/supplies/services by sub-contractors meet the specified requirement at source with regard to completed work quality in order to meet client's requirement as per the contract

To coordinate the implementation of the BCIM Management System through the implementation of the project quality plan in order to meet the company's internal policies and certification to international standards.

To liaise with construction team to ensure all inspection and testing requirements are fulfilled and completed work quality is achieved in order to meet client's requirement as per the contract.

To liaise with client's representative on site to ensure all inspection and testing requirements are fulfilled and complete work quality is achieved in order to meet client's requirement as per the contract.

Ensure that the quality system is fully established, implemented and maintained at all levels in line with the requirements of the Project Quality Plan

Implement and maintain a Management Information System that measures the effectiveness of the quality plan

Supervises the activities of the Laboratory personnel

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PLANNING INCHARGE

Reports to the Project Head.

To prepare construction schedule for the works with respect to baseline program

To prepare micro schedule for each activity and daily monitoring and reporting to the project manager

To prepare weekly and monthly look ahead program and convey the same to all department heads.

To find the critical activity details and convey it into the project manager

To coordinate the implementation of the BCIM Management System through the implementation of the project quality plan in order to meet the company's internal policies and certification to international standards.

To liaise with construction team to convey the daily planning and work schedule.

To liaise with client's representative on site to ensure daily reporting and project hindrances.

To monitor the daily progress and knowledge the site requirements to fulfill the activity completion through site managers.

Prepare weekly and monthly project reports and same to communicate the higher management and client.

To assist the project managers for smooth functioning of execution activities on a daily basis.

To make proper presentations for project activities time to time, if required.

Ensure that the materials quality control program is carried out as required in the Specifications and Project Quality Plan

Verify, confirm and certify conformance of the materials, finished product and testing requirements of the contract and the Project Quality Plan

Maintain a calibration schedule for all Lab testing equipment and ensure calibration is carried out as planned

QA/QC ENGINEER

As the Quality Management Representative responsible for the operation and effectiveness of quality management systems on Civil / Building Projects in accordance with local legislative requirements and companies policies and procedures.



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Responsible to control, review and maintain quality system documents within the Management System and at Project level. Also acts as controller for the distribution of all controlled documents associated with the Project / Management System.

Assists/advises the Office / Project staff in developing and implementing quality system and matters relating to the work requirements of the office / project.

Monitors the implementation of the quality systems and facilitates continual improvement initiatives to the systems.

Ensures all employees are aware of the quality requirements in their designated duties. Facilitates training and development of staff in quality management related fields.

Reports to the Quality in charge on quality management performance. Functional reporting of this performance to the Quality in charge.

Liaise with Client & Project Team to ensure all works are certified as accepted

STORE INCHARGE

Reports to the Project Manager

Manage the stores and track stores inventory to ensure availability of all construction materials

To assist to order consumables at minimum stock level IN ORDER TO maintain consumable stock and ensure consumable stock is in working condition.

To issue report and compile stock inventory inspection checks IN ORDER TO update on all consumable stock availability.

☐ Maintain proper records of all stores inventory. Reports to the project Manager in case of shortage of material. Maintain all materials, tools and equipment's and ensure they are in good condition at all times

Maintain material stocks to ensure a continuous supply of materials are available at all times and inform Section Manager / Buyer of any replacement required once the stock reached the minimum stock level

Ensure the materials, tools and equipment's in the store are always arranged neatly and locations of equipment maintained. Keep records and alert superiors of any abnormality in the consumption of materials, tools and equipment's

Keep records of all hired equipment's on site and ensure off-hire and breakdowns are recorded accordingly.

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Maintain User Daily Records of all materials, tools and equipment's being taken from the site store.

The responsibility and authority assigned to each member of the Project Team by their respective superior is documented in their Position Descriptions kept with the Human Resource Department.

3.1.2 Quality Responsibilities of Project Personnel

All project personnel are required to be familiar with the content of this management plan and to participate in implementing and maintaining the management system.

It is the responsibility of the respective operative functions to ensure that the requirements for quality are fulfilled for works under their responsibility.

All new staff and staff who are given new responsibilities are to be given induction into the requirements of this management plan in general and into their function and responsibilities in particular.

Responsibilities of other Key Personnel are defined in the Key Responsibility Matrix. Detailed responsibilities are set out in the Human Resource Department.

3.1.3 Quality Management Representative

The BCIM Quality in charge is the Quality Management Representative for the Project as defined in clause ISO 9001:2015.

The Quality Assurance in charge, in his role as the Management Representative, is responsible to liaise with the Client's / Employer's Representatives and Consultants on matters relating to the quality management system.

3.1.4 Reporting and Meetings

All Key Quality Team Personnel submit weekly reports to the Quality in charge. The format of these weekly reports is determined by the Quality in charge, based on the data required to enable him to analyze the performance of the quality system in the field and to compile his report for inclusion into the Monthly Project Report.

The Quality in charge calls for meetings at least once a month with his team as a forum to discuss problems encountered/anticipated and to disseminate critical information. Minutes are kept of the monthly meetings.

3.1.5 Facilities



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The Project Manager in conjunction with the Quality in charge ensures the Project Quality Team is provided with adequate facilities to perform their assigned tasks properly.

A laboratory facility will be built on site equipped with the necessary equipment required for testing.

Generally, all testing will be conducted on site. When testing frequency is very minimal and impractical, the tests will be conducted in an approved outside laboratory.

Where provision of such facilities is not possible or impractical, the Quality in charge out-sources the required services (e.g., laboratory) in conjunction with the Commercial Manager (Subcontracts/ Procurement).

3.1.6 Communication

Cell-phones /sim / wireless are provided to the Project Quality Team so that they can be contacted at all times during working works.

4. PROJECT CONSTRAINTS AND RISKS

Project Constraints	
Constraints	Controls
Prevailing attitude & Culture of Subcontractors	Robust inspection and testing Regime.
Suppliers towards quality	Extend awareness and training
	Programs to subcontractors / suppliers.
Limited coverage of QA/QC Engineers	Section Managers to be responsible for quality in their own packages.
	Implement frequent audits and ITP
	Verification on all Sections
Heavy Reliance on construction Team to Conduct day-to-day Inspections & Testing	Robust inspection and testing Regime
	Continuous awareness & Training

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	Program to construction team
	Implement frequent audits and ITP
	Verification on all Sections
Interfaces with the Client's /Employer's site representatives	Identify on ITP & Implements

4.1 Project Risks

Controls to manage the identified risks are documented in various sections in this and other related Project Plans and System documents. These are referenced in the Project Risk Identification matrices.

The Project Risk Identification matrices are restricted documents and for internal use only. Copies are held by the department In charges. No copies are to be issued to any external parties without prior permission from the Project Manager.

These matrices are reviewed and updated by the Project Manager monthly in conjunction with the relevant Manager and included in the Monthly Progress Report

5. QUALITY MANAGEMENT STRATEGY

Quality Planning

The Project Management System is developed specifically to meet the Client's expectations, based on the requirements of the Contract. Such requirements also include identification and analysis of risks and constraints.

Quality planning focus on:

Identified constraints and risks

Quality objectives and targets

Resources required

Competency and training needs

Develop and implement Quality Plan, ITP, MS as required



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Site plant & equipment maintenance, damage Investigation

Internal Audit to check compliance with Quality system

Meeting with Engineers / Contractors

Material Controls

Work conformance/non Conformance report during construction activities

Continual improvement

Quality toolbox induction and on job training for QC personnel

Maintaining Calibration register for measuring equipment

Maintaining daily inspection progress reports

Objectives and Targets

The Project Quality Team adopts the following for this Project:

Ensure operational works are carried out in an efficient and effective manner.	Develop, Implement and review BCIM Management System to suit Project Requirement in compliance with Legal and Statutory Requirements	Procedures/Work Method
Focus on Customer Satisfaction	Implement awareness of Quality Management	Statements/Inspection Test Plans are developed for Project activities100% System
Improve quality performance by reducing Quality Costs	System Timely quality delivery of the product	Induction is carried out Periodical Roll-Out of Management System Documents to the relevant team
Contractor and Supplier Management	Effective Control of Defects	With their relevant tasks. Adherence to the instruction
	Post-Performance evaluations is carried to all	The Summary Assessment to be updated
	Major Contractors and Suppliers	with Preferred list of Contractors and Suppliers
	Carry out the Quality Inspections/Audit for	

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Assist Management System	Contractors/ suppliers during the operation.	100% quality inspections & Audits carried out as per project inspection
Certification for participating Companies.	Identify root causes of the NC's and take actions to rectify the issues while prevent them from Recurring.	schedule/Supplier's Quality assurance plan / TPT100% Timely completion and taking over with complete set of hand over documents. No major NC raised by Certification Body.100% close out of identified NC's and all observations within time frame.

QUALITY MANAGEMENT SYSTEM

Document Control

Proper and timely control of documents contributes significantly to the successful management of a Project and to assuring quality of the works.

The Document Control systems' main objectives are to ensure that: -

Documents used are correct and up-to-date

Documents are recorded and kept systematically to allow easy retrieval and quick traceability.

Quality in charge is responsible for Document Management System implementation.

Project Quality QC & Systems shall appoint a document Controller to manage project documentation in soft and hard copies.

6.1.1 Quality System Documents Project Plans

Set out the structure and the content of the Management Systems, linking the policy and objectives with system elements and procedures

Set out the Project's organization, responsibilities and authorities



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Project Procedures

Provide detailed instructions as to what is to be done, by whom, when, where and how particular administrative processes are to be carried out.

Define how the process is to be controlled and what records are to be taken and maintained

Work Method Statements

Provide detailed instructions on how particular production or construction operations are to be carried out.

Standard Operating Procedure

Define how particular Project-specific administrative tasks are to be carried out

Are similar to Project Procedures in nature except that they are developed specially for the Project

Inspection and Test Plans (ITP)

Are related to Work Method Statements

Set out the responsibilities, inspection and testing methods, frequencies, acceptance criteria and verification records for defined work elements

Formats

Are records which demonstrate compliance with the requirements of a Procedure or Project Instruction?

Checklists

Are records which demonstrate compliance with the requirements of a Work Method Statement or

ITP

Conformance Reports / Works Report Sheets

Are documents of ITPs which provide records on inspection checks and tests on the works / materials as required under the ITP?

Conformance Reports certify discrete elements of the works meet contractual requirements

Reports

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Other reports than above such as test reports, audit reports, certificates, etc., related to fulfillment of quality requirements

Supplementary Documents

To achieve the project objectives defined in this document, the project may procure the services of various other contractors, suppliers and consultants. Each of these organizations shall be required to prepare, to a degree depending on the complexity of their particular service, quality plans such to document resources, practices and work methods. Such documents and related records will form part of the project quality system documentation.

Documents Type & Structure

Documents are roughly divided into the following categories. The descriptions provided are only an indication of the type of documents found and are by no means exhaustive.

Quality Systems Documents – as described above

Safety & Environmental Systems Documents – these are detailed in the “Safety Plan” and “Environmental Plan” respectively

Design & Technical Documents – e.g. Drawings, Specifications, Design Calculations.

Contractual Documents – e.g. Notification of Inspections, Interim Payment Certifications, Variation Orders, Claims

Correspondences

Site Administrative Documents – e.g. Inventories, Productivity Reports, Minutes of Meeting Office Administrative Documents – e.g. Statutory Permits, Licenses, Insurances, Personnel

Details, Payroll, Accounting documents

Commercial Documents – e.g. Subcontracts, Purchase Orders, Quotations, Invoices & Delivery Notes

Document Control Procedures

The below list out the Procedures which are relevant to Control of Documents: -

– Control of Correspondence

Material Control



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The project scope of work involves majority of services to be hired rather than procurement of materials.

The receiving, handling, storage, issuing and wastage control of material are guided by procurement department.

The following deals with the selection of vendors to ensure better services and right material supplied.

Pre-award Assessment

Prior to placement of a supply/contract agreement for material/services as potential suppliers are short-listed based on selection criteria established for the vendors. If required, the Commercial Manager (Subcontracts/Procurement) in conjunction with the *Quality Manager* is to plan visits to subcontractor/ supplier's premise to assess their capabilities, infrastructure facilities and resources to meet the required standard of quality.

Material Certification

All major material suppliers are required in their Agreements to provide conformance certificates or test reports confirming compliance of the material, traceable to the batches supplied. Proof of calibration for equipment used in the certification testing should also be included.

These certificates and/or test reports are forwarded to the Quality Manager for review. Any corrective and preventive action arising from the review is handled according to the Procedure.

Inspection of Procured materials

After award of Purchase order QA or QC engineer should coordinate with supplier and instruct for submission of Inspection and Test Plan (ITP) for review and approval by Quality Manager/Client TPI. Inspection of procured materials shall be conducted by QA or QC engineer or appointed TPI Inspector as per contract. The supplier shall send inspection notification within 15 days prior to inspection date.

Inspection Visits to Supplier's Premises

Periodic visits to the suppliers' production premises are planned and implemented by the Quality Manager. The purpose of these off-site visits to verify the checks and testing are being carried out correctly and to provide continuous confidence in the certification documents supplied.

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The frequency of these visits corresponds with the criticality of the material quality at certain work stages and standard of quality shown by the supplier as assessed by the Quality Manager

Design Process Control

To be developed

Construction and Production Control

In order to deliver the Project which successfully meets the Client's requirements, it is necessary to map out the processes or tasks and identify the inspections and / or testing required. Records demonstrating compliance of these activities have also to be defined. This is done through Inspection & Test Plans (ITPs)

Inspection and Test Plans

Inspection and Test Plans serve as a checklist specifying at what point of the process, certain inspection or/and test would be required, the frequency and the person responsible for the inspection and testing, acceptance criteria and type of records required.

All these planned inspection and testing are derived by correlating requirements from the Construction Program, Contract specifications, Work Method Statements, construction/shop drawings and industrial standards.

Inspection and Test Plans (ITP) are developed and controlled according to management system. It is required that all Supervisors responsible for the activities as per ITP to sign on the ITP to indicate having read and understood the requirements.

Appointed subcontractors are requested to submit their own ITP or QA Plan for review by the Construction Team with Quality Manager. Where the subcontractor is unable to produce ITP or where submitted ITP does not meet the requirements, the Planning Manager in consultation with Quality Manager decides whether to assist the subcontractor in amending the ITP or issue project-produced ITP for their use.

Any interfaces with the Client/TPI representatives with regard to joint and/or acceptance inspection must be included in the ITP to ensure these are carried out. HOLD points are included into the ITP only when it is required contractually. Additional hold points requested by the Client/TPI must be agreed with the Planning Manager.

A briefing is held whenever an ITP is issued for all engineers, supervisors and subcontractors involved to explain the usage of the ITP as well as to highlight any issues

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identified. This briefing is called by the Quality Manager and conducted by the originator of the ITP or other nominated person by the Quality Manager.

ITP Verifications

To ensure a most robust inspection and testing regime, verification is carried out to ensure the personnel responsible are implementing the planned inspection and testing so effectively and in timely manner.

The Construction Manager is responsible to identify the area and activity, plan, and assign personnel to carry out the verification activity. The frequency depends on the criticality of the activity, but generally, this is carried at least once a month for all Packages.

Site QC Engineers are nominated to undertake the verification activities as per approved ITP. Further hold or witness point activity related to Client TPI shall be intimated or notified prior to the activity commencement and to ensure their presence is there for necessary witness/hold point clearance as per approved ITP. The verifying site QC Engineers are also rotated with regards to the assigned activity or area.

Conformance Reports for each Lot are completed by Quality Manager or his delegate and submitted to the Construction Manager for review and coordinate any further action required. Lots, which are areas of work, are identified according to the main structures or work elements as identified on the Construction Drawings.

Laboratory Testing

While the Quality Manager has overall responsibility for laboratory testing, the Laboratory Manager is responsible for planning and carrying out the laboratory testing as required in the Contract Specifications and according to the relevant ITP.

All test results from the laboratory testing are submitted to the Quality Manager who then forwards them to the relevant Manager for any action necessary.

The Laboratory Manager is also responsible for ensuring that all equipment or external laboratories used for testing are calibrated and adequate. Verification visits are also carried out on any appointed external laboratories on a regular basis to provide continuous confidence in the reports submitted by these laboratories.

Testing Monitoring and Measuring of Equipment

Testing, Monitoring & Measuring Equipment are an important facet of any construction project as these ensure that the correct amount of materials is used or consumed, or they gauge the compliance of material and/or work element with the Specifications. A failure to ensure that these are in tip-top condition may result in losses for the organization due to non-compliance of works or excessive wastage. Testing, Monitoring and Measuring Equipment in this context does not refer to only the Laboratory Equipment, but also refer

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to Survey Instruments, Weighing Equipment (e.g. Weigh Bridge, Load Cells) and Measuring Equipment (e.g. Diesel Pumps).

Depending on the nature of the equipment, the Manager in charge identifies the equipment required and arranges for these to be procured. In the case of the Laboratory, the Laboratory Manager identifies all testing, monitoring and measuring equipment required by the Quality Engineers and Laboratory for verifying conformity of the Works to specifications and arranges for them to be procured.

Most of this equipment are sensitive, and require periodic verification of its accuracy through Calibration.

Competency and Training of Manpower

In addition to the formal performance appraisal system and subsequent training needs identification, the Construction Manager with Quality Manager also assess on-the-job the competency of his team members. The Quality Manager determines the type of training required and arranges for the training as and when planned by divisional HR section. Such training can be in the form of external, internal or on-the-job courses.

As minimum, the following types of training must be completed.

Topics	Type of training (as a minimum)
Management System	Induction
Usage of Inspection & Test Plans	Briefing
Daily work briefing	Briefing/Toolbox meeting
Material Handling – Good Practices	Briefing/Toolbox meeting
Awareness of ISO 9001:2008 requirements	Internal/External source
Work procedures	Training workshop Internal
Internal Quality Auditing	Training workshop Internal

All trainings are managed according to Corporate Procedure.

The first 4 training topics listed above are also extended to the subcontractors responsible for the various trades identified. The Quality Manager in conjunction with the Construction Manager ensures the subcontractors attend these trainings.

Non Conformance Control

Non-conformance are identified, reported and rectified according to project Procedure.



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Coordination and maintaining an actual potential quality non-conformance control system including the issuing of NCR reports in case of non-conforming products and ensure corrective and preventive actions are implemented effectively. Raising Non Conformance Reports (NCR) is not solely by the Quality Team. The Project Construction Team is also responsible to raise NCR.

The Quality Manager is responsible to perform the analysis of non-conformances and include the analysis into his Monthly Project Report.

System Performance Measurement

Auditing

The implementation and effectiveness of the BCIM Management System is verified by conducting periodic audits.

The Quality Manager prepares and maintains the Level 1 internal Audit and inspection Schedule to define the extent and type of audits during the execution of project. The schedule is reviewed and updated by Project Manager as & when necessary to ensure it remains current.

Audits are conducted & reported and subsequent improvement actions addressed for necessary action. The Quality Manager nominates the auditor(s) for those audits he scheduled for the Project.

Upon completion of the audit reports, the Quality Manager distributes copies to:

Project Director

Managing Director

Data Analyses

Various data are collected and analyzed as indicators of the performance of the quality management system. Such data includes the following.

Objectives and targets for the Quality Dept. Audit findings.

Non Conformance

The Quality Manager collates the data collected from the various sources as set out in the Procedures and coordinates with the department as action arising from these analyses. These are reported in the Monthly Project Report.

The above list does not include technical data for measuring quality level of finished work; like concrete cube results, soil compaction results, material testing, etc. These are tracked and compiled separately as per the relevant ITP.

Continual Improvement

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Management Reviews

As the Project runs on an integrated system, review of the quality system is carried out as part of the review of the overall Project Management System.

Reviews on the on-going Quality Control issue are also carried in the monthly quality meeting with the Quality Team.

Corrective and Preventive Actions

Project Procedure and *Corrective and Preventive Actions* provides the instructions for how actions arising from the reviews to be dispositional.

The Quality Manager is responsible to coordinate with the relevant manager(s) to ensure the action are closed and documented. Such documented corrective and preventive actions are included into the *End of Project Report* for the benefit of future projects.

Business Unit Quality Meetings

To be scheduled and maintained by the MD/CEO – Management Systems as required to meet the projects and Business Unit needs.

Lesson Learned

Construction Manager and Quality Manager are responsible to develop and maintain a

Lessons Learned Schedule to ensure that all 'lessons Learned' is captured as and when they occur and when they are fresh in the minds of the Project personnel. The lessons learned review meeting shall be conducted at planned intervals to review all encountered problems as well as good practices for continual improvement of BCIM management system.

Appendix

Appendix A – Work Method Statements and ITP

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